STAT 155 - HW 1

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1 Problem 1.1

1.1 (a) System Description

The system I will analyze is the modern academic research ecosystem. This system is composed of universities, funding agencies, and academic journals. Its primary goal, from a societal perspective, is to advance human knowledge and provide high-quality education. The system's designers—university administrators, governments, and funding bodies—use performance metrics to allocate resources, grant tenure, and award prestige.

The core incentive problem arises from what the system claims to value versus what it actually rewards. The stated goals are high-quality research, impactful discoveries, and excellent teaching. However, the primary metrics for career advancement are the *quantity* of publications and the prestige of the journals they appear in. This creates a fundamental misalignment between the designer's goal (long-term knowledge creation) and the participant's goal (short-term career advancement). Researchers are incentivized to maximize their publication metrics, even if it comes at the expense of research quality or teaching effectiveness.

1.2 (b) Evidence of System Gaming

There is significant evidence that researchers strategically game this system to advance their careers. This manipulative behavior includes:

- **"Salami Slicing":** This is the practice of dividing a single research study into the largest possible number of "least publishable units." Instead of a single, comprehensive paper, a researcher might publish several smaller, less impactful articles to increase their publication count.
- **Citation Manipulation:** To boost their perceived impact, some researchers form "citation clubs" where they agree to cite each other's work, regardless of relevance. This artificially inflates their citation metrics, which are a key measure of influence.
- **HARKing (Hypothesizing After the Results are Known):** This is a practice where researchers conduct a study, find a statistically significant correlation, and then write the introduction to pretend that this finding was their original hypothesis. This makes the results seem more important and increases the chances of publication, but it undermines the scientific method and contributes to a "reproducibility crisis" where other scientists cannot replicate the findings.
- **Neglect of Teaching:** Because promotions and funding are heavily tied to research output, many professors strategically minimize the time and effort they put into teaching. This leads to a decline in educational quality, as the reward structure disincentivizes a core function of the university.

1.3 (c) Proposed Improvements

To mitigate these incentive problems, the system's evaluation criteria should be redesigned to align with its original goals. My proposed solution is to shift from a quantity-based model to a multidimensional, quality-focused evaluation framework.

1. **Implement a "Portfolio Review" Model:** Instead of counting publications, tenure and promotion committees should require researchers to submit a small, curated portfolio of their most significant work (e.g., their three best papers). The researcher would include a narrative explaining the work's impact and importance. This would incentivize researchers to focus on producing high-quality, impactful work rather than a high volume of mediocre papers. It directly counters "salami slicing" by making a single, comprehensive study more valuable than many small ones.

2. **Diversify and Reward Other Contributions:** The evaluation system must formally recognize and reward other essential academic activities. This includes peer review, mentoring junior researchers, developing new courses, sharing data sets for public use, and engaging in science communication. By rewarding these "academic citizenship" activities, the system would encourage a more holistic and collaborative research environment, reducing the singular focus on individual publication metrics.

3. **Reform Teaching Evaluation:** To address the neglect of teaching, evaluations must go beyond simple student satisfaction surveys. Incorporating peer reviews of teaching, assessments of student learning outcomes, and rewards for pedagogical innovation would create a tangible incentive for professors to invest in their educational role.

This revised system would reduce strategic behavior by making it less effective. If publication count is no longer the primary metric, then gaming it through "salami slicing" or citation clubs yields diminishing returns. By aligning the reward structure with a broader set of valued contributions—high-quality research, robust teaching, and collaborative citizenship—the system would encourage participants to pursue the actual goals of academia, not just their proxies.

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2 Exercise 2.3

2.1 Mechanism.

Collect bids $b_1, \ldots, b_n$. Rank them in nonincreasing order $b_{(1)} \geq b_{(2)} \geq \cdots \geq b_{(n)}$. Allocate the k identical items to the k highest bidders (the bidders who placed $b_{(1)}, \ldots, b_{(k)}$). Each winning bidder pays the $(k+1)$ -th highest bid $b_{(k+1)}$ (if a bidder does not win they pay 0). Tie-breaking should be done by a rule that does not depend on any single bidder's report (for example, by a fixed public ordering over bidders or by an independent public coin).

This is the natural multi-unit analogue of the second-price auction (it is the Vickrey / VCG auction for unit demand).

2.2 Proof that truthful bidding is a dominant strategy (DSIC).

Fix any bidder i. Let the other bidders' bids be $c_1 \geq c_2 \geq \cdots \geq c_{n-1}$ (the ordered bids of the $n-1$ other agents). Define the *threshold*

$$t := c_{(k)},$$

the k -th highest bid *among the other bidders*. (Because $n > k$, $c_{(k)}$ exists.)

Consider bidder i with true valuation v_i for one unit. Compare outcomes when i bids some b_i to the outcome when i bids truthfully $b_i' = v_i$. The mechanism's allocation rule relative to the fixed other bids is simple:

- If $b_i \geq t$, then i becomes one of the top k bidders and **wins** one item.
- If $b_i < t$, then i **loses**.

When i wins, the payment in this mechanism is the $(k+1)$ -th highest overall bid. Since a winning bidder is among the top k, the $(k+1)$ -th overall equals t (it is determined entirely by the other bids). Thus the payment of i when they win is exactly t, which does **not** depend on i 's own bid (given other bids fixed).

So i 's utility when bidding b_i is

$$u_i(b_i) = \begin{cases} v_i - t & \text{if } b_i \geq t, \\ 0 & \text{if } b_i < t. \end{cases}$$

Now compare truthful bidding $b_i' = v_i$ with any deviation b_i:

1. If $v_i \geq t$: truthful bidding wins and yields utility $v_i - t \geq 0$. Any higher bid still wins but payment remains t, so utility unchanged. Any lower bid that falls below t causes loss and utility 0, which is weakly worse. Hence bidding v_i weakly dominates all deviations.

2. If $v_i < t$: truthful bidding loses and yields utility 0. Any deviation that still bids below t leaves utility 0. Any deviation that raises the bid to $\geq t$ would cause a win but a payment of $t > v_i$, giving utility $v_i - t < 0$, strictly worse. Hence bidding v_i is strictly better than any deviation that would make the bidder win.

Thus for every possible profile of other bids, truthful bidding maximizes i 's payoff (weakly), so reporting the true valuation is a dominant strategy. Because this holds for every bidder, the mechanism is DSIC.

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3 Exercise 2.5

3.1 Mechanism

Each contractor i reports a cost b_i (a nonnegative number). Let b_{-i} denote the vector of others' reports and write

$$m := \min_{k \neq i} b_k$$

for the minimum of the other bids (for statements below, it is useful to also write $b_{(1)}$ for the smallest bid overall and $b_{(2)}$ for the second-smallest).

1. **Allocation rule:** Award the contract to the bidder with the smallest reported cost $b_{(1)}$. (If there is a tie for the minimum, break ties by a fixed deterministic rule or uniformly at random among the tied lowest bidders.)

2. **Payments:** If bidder i loses, pay 0. If bidder j wins (so $b_j = b_{(1)}$), pay the winner

$$p_j \;=\; b_{(2)} \;=\; \min_{k \neq j} b_k.$$

(So the winner is paid the second-smallest reported cost; everyone else gets zero.)

3.2 Properties

1. **Selection:** If all contractors report truthfully $b_i = c_i$ (their true costs), the mechanism selects the contractor with smallest true cost (the smallest c_i), because the allocation just picks the bidder with the smallest reported cost.

2. **Winner's payment $\geq$ reported cost:** Under truthful reports, the winner's payment equals the second-smallest true cost $c_{(2)}$, which is $\geq c_{(1)}$, so the payment to the winner is at least her reported cost. Losers are paid nothing by construction.

3. **Truthfulness (dominant-strategy incentive compatibility):** show truthful reporting dominates any misreport for any fixed reports of others.

Fix bidder i with true cost c_i. Fix others' reports and let $m = \min_{k \neq i} b_k$.

- If $b_i > m$ then i loses and gets utility 0.
- If $b_i < m$ then i wins and gets utility $u = p_i - c_i = m - c_i$ (note payment equals m, which depends only on others' bids, not b_i, as long as $b_i < m$).

So as a function of b_i (with b_{-i} fixed) the utility is

$$u(b_i) = \begin{cases} 0 & \text{if } b_i \geq m, \\ m - c_i & \text{if } b_i < m. \end{cases}$$

There are three cases for c_i:

- If $c_i < m$: then $m - c_i > 0$. Any report $b_i < m$ yields the same positive utility $m - c_i$; any report $b_i \geq m$ yields 0. In particular reporting $b_i = c_i$ (which is $< m$) achieves the optimal utility.
- If $c_i > m$: then $m - c_i < 0$. Winning would give negative utility, so the best thing is to lose (utility 0). Any report $b_i \geq m$ ensures losing; reporting truthfully $b_i = c_i$ (which is $\geq m$) yields the optimal utility 0.
- If $c_i = m$: the bidder is indifferent; truthful reporting achieves the maximum utility (ties aside).

Because for every possible profile of other bids b_{-i} the bidder maximizes utility by reporting $b_i = c_i$, truthful reporting is a dominant strategy.

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4 Exercise 2.8

4.1 Lemma

If $a_1 \geq a_2 \geq \cdots \geq a_n$ and $b_1 \geq b_2 \geq \cdots \geq b_n$, then for any permutation π of $\{1, 2, \ldots, n\}$,

$$\sum_{i=1}^{n} a_i b_i \geq \sum_{i=1}^{n} a_i b_{\pi(i)}.$$

Proof of Lemma:

- **Base case ($n = 2$):** Given $a_1 \geq a_2$ and $b_1 \geq b_2$,

$$a_1 b_1 + a_2 b_2 - (a_1 b_2 + a_2 b_1) = a_1(b_1 - b_2) - a_2(b_1 - b_2) = (a_1 - a_2)(b_1 - b_2) \geq 0,$$

  so $a_1 b_1 + a_2 b_2 \geq a_1 b_2 + a_2 b_1$.
- **Inductive step:** Assume the lemma holds for $n - 1$. Consider any permutation π for n elements.
  - If $\pi(1) = 1$, then the sum includes $a_1 b_1$. The remaining terms involve $a_2, \ldots, a_n$ and $b_2, \ldots, b_n$, and by the induction hypothesis,
$$\sum_{i=2}^{n} a_i b_i \geq \sum_{i=2}^{n} a_i b_{\pi(i)},$$
    so the overall inequality holds.
  - If $\pi(1) \neq 1$, let $\pi(1) = k$ (where $k \neq 1$) and let j be such that $\pi(j) = 1$. Since $a_1 \geq a_j$ and $b_1 \geq b_k$ (as b_1 is the largest),
$$a_1 b_1 + a_j b_k \geq a_1 b_k + a_j b_1$$
    by the base case. Thus, swapping the assignments to pair a_1 with b_1 and a_j with b_k does not decrease the sum. After this swap, we are in the case where $\pi(1) = 1$, and the result follows by induction.

Hence, the lemma is proved.

4.2 Apply the Lemma to Sponsored Search

Assume (WLOG) $v_1 \geq v_2 \geq \cdots \geq v_n$ (we relabel bidders so valuations are sorted).

Let S be any set of k bidders assigned to the slots. Let $v^S_{(1)} \geq v^S_{(2)} \geq \cdots \geq v^S_{(k)}$ be the valuations of the bidders in S sorted in descending order.

By the lemma, for fixed S, the maximum social welfare achievable by assigning S to the slots is obtained by pairing the highest valuation in S with the highest CTR, the second highest with the second highest CTR, etc. That is,

$$\max_{\text{assignments of } S} \sum_{i \in S} v_i \alpha_{j(i)} = \sum_{i=1}^{k} v^S_{(i)} \alpha_i.$$

Now, since $v_1, v_2, \ldots, v_n$ are sorted, for any set S of k bidders, the i-th highest valuation in S is at most the i-th highest valuation overall:

$$v^S_{(i)} \leq v_i \quad \text{for all } i = 1, \ldots, k.$$

Because the top k valuations overall are $v_1, \ldots, v_k$, and any other set S must have valuations that are element-wise dominated.

Since $\alpha_i > 0$ for all i,

$$\sum_{i=1}^{k} v^S_{(i)} \alpha_i \leq \sum_{i=1}^{k} v_i \alpha_i.$$

Thus, for any set S, the maximum social welfare achievable is at most $\sum_{i=1}^{k} v_i \alpha_i$.

Moreover, if assign the top k bidders (with valuations $v_1, \ldots, v_k$) to the slots in order (i.e., bidder with v_i to slot with α_i), we achieve exactly

$$\sum_{i=1}^{k} v_i \alpha_i.$$

Therefore, this assignment maximizes social welfare.

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5 Problem 2.1

5.1 (a) The auction form is DSIC

Fix an arrival index i. By the model the price p_i is a function of earlier bids only:

$$p_i = f_i(b_1, \ldots, b_{i-1})$$

so it does not depend on bidder i 's own report b_i.

For bidder i two cases:

- If $v_i \geq p_i$: by bidding any $b_i \geq p_i$ the bidder wins and pays p_i, giving utility $v_i - p_i \geq 0$. Bidding $b_i < p_i$ makes the bidder lose with utility 0. So some bid $\geq p_i$ (in particular the truthful $b_i = v_i$) weakly dominates any bid $< p_i$.
- If $v_i < p_i$: winning would yield negative utility, so the bidder's best action is to ensure losing, i.e. bid $b_i < p_i$. Bidding $b_i = v_i$ (truthful) causes losing, which is optimal.

Thus for every possible p_i the bidder's utility is maximized (weakly) by bidding truthfully $b_i = v_i$. Hence the auction is DSIC.

5.2 (b) No deterministic online auction gives a constant fraction of the optimum for arbitrary arrival order

Fix a deterministic online algorithm A. Consider the following family of valuation profiles parameterized by a large number $M > 0$. Let every bidder except one have valuation 1. Put a single "special" bidder whose valuation is M (so max valuation is M).

Because A is deterministic, its prices and selling decision up to some position depend only on the observed bids (which will be 1 s) and thus are fixed when the history consists solely of ones. Let s be the earliest index (possibly $s = \infty$ if it never sells) at which A would sell when it has seen only bids equal to 1 so far.

Now the adversary chooses the arrival order as follows:

- If s is finite, place the M -bidder **after** index s. Then on the instance where the huge bidder is placed after s, algorithm A will already have sold (by index s) to a bidder of value 1, giving welfare at most 1, while the optimal welfare is M. For any fixed $c > 0$ pick $M > 1/c$ and the ratio $1/M < c$, so A fails.
- If $s = \infty$ (i.e. A never sells when it keeps seeing ones), then place the M -bidder at any position. When M arrives the price p posted by A is determined by the history of ones; either $p \leq M$ and A sells to the M -bidder (good), or $p > M$ and A rejects M and ends with welfare at most 1. In the latter case choose $M > 1/c$ and again the ratio $\leq 1/M < c$, so A fails.

In either case the adversary (who knows A) can place the high-value bidder so that the welfare achieved by A is at most 1 while the optimum is M. Since we can make M arbitrarily large, no deterministic algorithm A can guarantee a fixed constant fraction $c > 0$ for all orders and valuations. This proves the claim.

5.3 (c) A randomized-order algorithm with constant guarantee — show $c = \frac{1}{4}$ works

Use the suggested algorithm (a simple "sample-then-threshold" rule):

- Reject (do not sell to) the first $\lfloor n/2 \rfloor$ arrivals, but record their bids.
- Let p be the maximum bid observed among the first $\lfloor n/2 \rfloor$ bidders.
- Among the remaining $\lceil n/2 \rceil$ bidders, sell to the first bidder whose value $\geq p$ (if any), at price p. If none meets the threshold, no sale.

Assume truthful bidding. Let $v_{(1)}$ be the highest valuation and $v_{(2)}$ the second-highest. Consider the event

$$E = \{\ v_{(1)} \text{ appears in the second half and } v_{(2)} \text{ appears in the first half }\}.$$

If event E occurs then the maximum among the first half equals $v_{(2)}$ (so $p = v_{(2)}$), and when $v_{(1)}$ arrives in the second half we have $v_{(1)} \geq p$ so the algorithm will accept $v_{(1)}$. Hence conditioned on E, the auction obtains welfare exactly $v_{(1)}$ (the optimum).

So the expected welfare is at least

$$\Pr(E) \cdot v_{(1)}.$$

We need a lower bound on $\Pr(E)$. Order is uniform over all permutations. Count ordered pairs of positions for the top two values: there are $n(n-1)$ ordered placements for (position of $v_{(1)}$, position of $v_{(2)}$). The number of favorable ordered placements where $v_{(1)}$ is in the last $n/2$ and $v_{(2)}$ is in the first $n/2$ is exactly $(n/2)\cdot(n/2) = n^2/4$ (assume n even for simplicity; for odd n the same asymptotic bound holds and constants can be adjusted).

Thus

$$\Pr(E) = \frac{(n/2)^2}{n(n-1)} = \frac{n}{4(n-1)}.$$

For every $n \geq 2$ we have $\dfrac{n}{4(n-1)} \geq \dfrac{1}{4} \cdot \dfrac{n}{n-1} \geq \dfrac{1}{4}$ (indeed the quantity tends to $1/4$ as $n \to \infty$ and is larger for small n). Therefore $\Pr(E) \geq 1/4$ for all n.

Consequently the algorithm's expected welfare is at least

$$\Pr(E) \cdot v_{(1)} \geq \tfrac{1}{4}\, v_{(1)}.$$

So the algorithm is a deterministic rule (the algorithm itself is deterministic) whose performance against a uniformly random order achieves expected welfare at least $c = \frac{1}{4}$ times the maximum valuation.